

ARYAN MITTAL

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EDUCATION

Georgia Institute of Technology

Atlanta, GA

Master of Science in Computer Science: Machine Learning, GPA: 4.0

Expected May 2026

- Planned Coursework: Statistical Learning, Machine Learning Theory, Stochastic Processes, Convex Optimization

Georgia Institute of Technology

Atlanta, GA

Bachelor of Science in Computer Science, Minor in Mathematics, GPA: 4.0

May 2025

- Relevant Coursework: Deep Learning, Machine Learning, Probability & Statistical Theory, Information Theory, Game Theory, Automata & Complexity Theory

EXPERIENCE

Millennium Management

June 2025 — August 2025

Quantitative Software Engineering Intern

Miami, FL

- Developing question-answering model for firm risk guideline system, enabling complex reasoning regarding Greeks, exposures, Value at Risk (VaR), and stress test metric trends by portfolio and risk managers

Microsoft

May 2024 — August 2024

Software Engineering Intern: Azure Core Networking

Atlanta, GA

- Decreased network node failure detection time from **72 hours** to **1 hour** and saved **8+** maintenance hours weekly by designing and implementing .NET service in C# to continuously monitor the health of **2,000+** routers
- Deployed service to **198** Azure servers **globally** and improved security by eliminating **3,000** yearly remote server logins
- Created utility service to export **16,000** rows of test result data per hour using Kusto queued ingestion API
- Authored comprehensive documentation on both services for use by team of **30+** developers
- Skills:** C#, .NET Framework, Computer Networking, Azure, Kusto, Azure Data Explorer, PowerShell

Data Science Student Researcher

January 2023 — Present

Joel Sokol Lab (Georgia Tech)

Atlanta, GA

- Contracted by **MLB** team to enhance league parity by developing mixed-integer **optimization model** for division realignment and expansion team location selection
- Backtested model on 30 years of historical league results resulting in average **22%** increase in division quality
- First-authoring 2 publications** on model and presenting results to MLB Commissioner's Office
- Designed logistic regression Markov chain model to rank NFL teams and predict playoff outcomes with **63%** accuracy (outperforms Vegas spread)
- Skills:** Python, PyTorch, Scikit-learn, Pandas, NumPy, Gurobi, Linear Programming, Nonlinear Optimization

UPS Supply Chain Solutions

June 2023 — August 2023

Software Engineering and Analytics Intern

Alpharetta, GA

- Prototyped **98%** accurate cloud-based computer vision application for package damage detection and reporting (est. annual savings **\$5M**)
- Developed **Weibull regression** model to estimate task completion times to within **2** business days
- Corrected pay rates for **350K** drivers by restructuring timecard database (**1.5M** entries) using Python script
- Skills:** Power BI, Google Cloud Platform, BigQuery, Vertex AI, Python, Pandas, Flask, Java, Azure DevOps

Mathematics Teaching Assistant

August 2023 — May 2025

School of Mathematics, Georgia Tech

Atlanta, GA

- Teach weekly studio/recitation sessions to **70+** students in Linear Algebra, Multivariable Calculus, and Calculus I

RESEARCH AND PROJECTS

The Cordiality Game and the Game Cordiality Number | thearyanmittal.com/projects/cordiality

- Invented graph-theoretic game with applications in statistical physics and proved **2** optimal play theorems
- Authored final paper **published** in *Graphs & Combinatorics*

Predicting Momentum Swings at Wimbledon | thearyanmittal.com/projects/mcm

- Developed **Markov chain** model to capture shifts in competitive advantage across 7000+ Wimbledon plays
- Trained **XGBoost** classifier to predict pivotal points and evaluated model sensitivity with **ablation study**
- Awarded **Top 20%** (10,000+ teams) in the 2024 COMAP MCM for 100-hour quantitative research challenge

Portal Laptop | thearyanmittal.com/projects/portal

- Built **\$300** prototype laptop optimized for **20ms** average remote desktop latency with **sub-30 second** user setup time
- Led **178** user interviews, conducted pricing analysis, and performed full statistical evaluation, visualization, and **hypothesis tests** of latency/jitter/packet loss experiments using **mixed effects models**
- Designed and engineered **secure** backend in **Rust** for auth system, peer-to-peer routing, and device management